

# Tuples: Immutable Data for Safe Trading

## What Are Tuples?

Tuples are similar to lists with one critical difference: once created, they cannot be modified. This immutability makes tuples perfect for data that shouldn't change, like:

- Historical trade records (date, price, volume)
- Order parameters (symbol, order type, quantity, price)
- Risk limits (maximum position size, stop-loss percentage)
- Market session times (open, close)

## Creating Tuples: Fixed Data Collections

Creating a tuple is similar to creating a list, but uses parentheses instead of square brackets:

```
1. # A tuple of daily price data (open, high, low, close)
2. daily_ohlc = (142.56, 145.32, 142.08, 143.78)
3.
4. # A tuple of trade information
5. trade_record = ("AAPL", "BUY", 100, 143.78, "2023-07-21")
6.
7. # A tuple with a single element (note the comma)
8. max_position = (1000,)
9.
```

Note that for a single-element tuple, you need a trailing comma to distinguish it from a simple parenthesized expression.

## Accessing Tuple Elements: Similar to Lists

Accessing elements in a tuple works exactly like lists, using zero-based indexing: